

GRUHESH SRI SAI KARTHIK KURRA

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AI/ML research engineer who builds systems from the math up — transformers, diffusion, and attention re-implemented from scratch, then shipped to production and on-device. Published in IEEE ICCMMLA 2025; independent research on neural weight compression and deepfake forensics. Incoming MSc Computing (AI & ML), Imperial College London; B.Tech CSE, 9.72/10 CGPA.

EDUCATION

Imperial College London

London, UK

MSc Computing (Artificial Intelligence and Machine Learning) | Incoming Sep 2026 – Sep 2027

- Unconditional offer accepted; CID 06079515. Focus: NLP, deep learning, probabilistic inference.

KL University

Hyderabad, India

Bachelor of Technology in Computer Science and Engineering | CGPA: 9.72/10 Aug 2021 – May 2025

- English proficiency: IELTS Academic 7.5 · Cambridge C1 Advanced.

EXPERIENCE

AI Integration Engineer

Oct 2025 – Nov 2025

InfoAjax Consulting Pvt Ltd

Remote

- Led development of an AI-driven integration framework connecting enterprise applications with cybersecurity services through standardized APIs and intelligent orchestration.
- Architected an agent-based system using Azure OpenAI for automated schema mapping and adaptive connection strategies that learn from historical patterns.
- Built a full-stack app (React 18, Vite, TailwindCSS, FastAPI) with WebSocket integration for real-time monitoring and agent coordination.
- Integrated Azure Active Directory via Microsoft Graph API for application discovery and automated metadata management.

WSO2 API Developer

Oct 2024 – Jun 2025

InfoAjax Consulting Pvt Ltd

Remote

- Developed and deployed RESTful APIs for PLDT (Philippines' largest telecom) using WSO2 Integration Studio for ticketing-system integration.
- Managed production ticket systems with secure credential handling via GitLab and Choreo cloud hosting with vault-based secrets management.
- Debugged systematically in an environment with sparse WSO2 documentation; engaged international clients directly under strict SLAs.

Founder & CEO

Mar 2024 – Jan 2025

Zyntheticx — Personal Research Venture

Remote

- Designed and built a synthetic-data generation platform producing privacy-preserving tabular, image, and text datasets with GANs and VAEs to address dataset scarcity.
- Prototyped a hierarchical meta-controller that coordinates specialized generator models to produce large, de-duplicated datasets, paired with a transformer-based PII detection and masking layer.
- Led a 3-developer team; established Git workflows and CI/CD.

Research Intern

Dec 2024 – Jun 2025

IIIT Hyderabad

Hyderabad, India

- Trained in LLM optimization — quantization, pruning, and knowledge distillation.
- Led a Layout-Preserving Document Translation system with a Vision-Transformer layout detector (92% accuracy) and a FastAPI backend integrating Tesseract OCR for multi-language extraction.
- Presented at the IIIT Hyderabad research expo (May 2025) with live demos to faculty and industry visitors.

AI & ML Research Trainee

May 2024 – Oct 2024

iHub-Data, IIIT Hyderabad

Hyderabad, India

- Completed a 6-month apprenticeship in AI/ML research methodology: LLM fundamentals, prompt engineering, fine-tuning, quantization, and deployment.

Data Science Intern

Jul 2023 – Oct 2023

Xtraleap India Pvt Ltd

Remote

- Built end-to-end ML pipelines — EDA, statistical modeling, data preprocessing, feature engineering, and model evaluation with pandas, NumPy, and scikit-learn.

PROJECTS

MANCE: Morphology-Aware Nested Character Embeddings | *PyTorch, NLP* [Code](#)

- Word-representation framework modeling words as nested character sequences (n-grams through full words), fused hierarchically with character-level Transformers, CNNs, and BiLSTMs.
- +58% accuracy on morphologically-varied sentiment (91.67% vs 33.33% baseline), 99.0% on DBpedia; outsized gains on German NER (GermEval 2014). Accepted at IEEE ICCCMMLA 2025.

DeepGuard — iOS On-Device Deepfake Detector | *Swift, Core ML, FastAPI* [Code](#) | [Demo](#)

- EfficientNet-B1 exported to a 26MB Core ML model running fully on-device at <500ms inference; 98.62% accuracy on 25k images.

BIE: Bit-Index Encoding for Weight Compression | *PyTorch, NumPy, Numba* [Code](#) | [Zenodo](#)

- 40× compression at 95% sparsity with MSE < 10⁻⁶; three encoding variants JIT-optimized with Numba, benchmarked across 253 configurations.

Hybrid RAG-Enhanced Deepfake Detection | *RAG, FAISS, CNN* [Zenodo](#)

- Privacy-first detector combining RAG with visual anomaly detection; 94.8% accuracy, 94.6% F1.

Large-Scale Models From Scratch — GPT-2 & NL2SQL | *PyTorch* [GPT-2](#) | [NL2SQL](#)

- Implemented GPT-2 (124M) from scratch — custom BPE tokenizer, multi-head attention, full training pipeline. Seq2seq NL→SQL; perplexity 1.42.

From-Scratch Collection — 25+ algorithms | *NumPy, PyTorch* [GitHub](#)

- Transformers, Diffusion (DDPM), GANs, VAEs; GNNs (GCN 79.3%, GraphSAGE 76.8%, GAT 81.9%); RL (TD, SARSA, Q-learning); classic ML without sklearn.

PUBLICATIONS

- **Morphology-Aware Nested Character Embeddings for Word Representation**, IEEE ICCCMMLA 2025 (7th). [IEEE Xplore](#)
- **BIE: Bit-Index Encoding**, Zenodo 2025, DOI: 10.5281/zenodo.17217218. [Link](#)
- **Hybrid RAG-Enhanced Deepfake Detection**, Zenodo 2024, DOI: 10.5281/zenodo.16732053. [Link](#)
- **Dynamic Auto-Finetuning of Language Models**, IJNRD vol. 9(11), 2024.
- **Global Remote RAM Sharing Framework**, IJNRD vol. 9(12), 2024.
- **Voice-Activated AI for Seamless Computer Interaction**, IJNRD vol. 10(1), 2025.
- **RADAR: Reasoning-Augmented Deepfake Artifact Recognition** (preprint, 2026). Multi-branch forensics architecture with iterative cross-attention reasoning.
- **Dual-Stream Artifact Detection with Iterative Evidence Refinement** (preprint, 2026). 96.3% AUC, 227k frames, public code + corpus. [Link](#)
- **Prism Tuning: Repulsion-Trained Seed Embeddings for Non-Redundant Generation** (preprint, 2026). Trainable diversity for LLM synthetic-data generation at zero inference overhead.

TECHNICAL SKILLS

Primary: Python, PyTorch, Hugging Face Transformers, NumPy, Azure OpenAI, FastAPI, Docker, Git

Specializations: LLMs; Generative (GANs, VAEs, Diffusion); Transformers; GNNs; model compression; RAG; Vision/NLP; synthetic data

Secondary: TensorFlow, LangChain, scikit-learn, Core ML; React, Next.js, Node.js, TailwindCSS

Languages: Python, R, JavaScript, TypeScript, Java, C, C++, SQL

Cloud & DevOps: AWS, Azure, GCP, GitLab, CI/CD, WSO2

Databases: PostgreSQL, MongoDB, MySQL, Supabase, Firebase, Redis

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate (2023) · AWS Cloud Practitioner (2023)
- Google TensorFlow Developer (2024) · Oracle OCI Generative AI Professional (2024)
- WSO2 Micro Integrator Practitioner & Developer V4 (2025)

AWARDS & LEADERSHIP

- **1st Place** — **University Webathon**, KL University (2022); **2nd Place** — **Design Expo** (2022–23).
- **Technical Lead, Student Activity Council**, KL University (2023–25); SAC Momentum Award.
- **National Tennis Ranking 395 (India)**, AITA (2014–19).